



Midterm 161122 statmeth

Statistical methods for AI (Vrije Universiteit Amsterdam)



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Midterm Exam Statistical Methods

Vrije Universiteit Amsterdam, Faculty of Exact Sciences

9.00 – 10.30h, November 22, 2016

- Question 1 is on this page.
- Always motivate your answers.
- Write your answers in English.
- Only the use of a simple, non-graphical calculator is allowed.
- Programmable/graphical calculators, laptops, tablets, e-readers, mobile phones, smartphones, smartwatches, books, own formula sheets, etc. are not allowed.
- On the last two pages of the exam, some formulas and tables that you may want to use can be found.
- The total number of points you can receive is 45: $\text{Grade} = 1 + \frac{\text{points}}{5}$.
- The division of points per question and subparts is as follows:

Question	1	2	3	4	5
Part a)	2	2	4	3	1
Part b)	2	3	4	2	2
Part c)	2	3	2	5	3
Part d)	2	-	-	3	-
Total	8	8	10	13	6

1. Are the following statements correct or incorrect? Choose one of the two options for each statement. Briefly motivate your answer.
 - a) According to the “68–95–99.7” rule, about 95% of the data points in a sample originating from a normal distribution will fall within 3 standard deviations of the population mean.
 - b) The sample mean and the sample mode are always equal for data at the ordinal level of measurement.
 - c) The sample standard deviation cannot be computed if the sample size n equals 1.
 - d) The probability of a union of two events is always less than or equal to the sum of the probabilities of these two events.

2. Executives of a news portal investigate the amount of time its visitors spend on the main page of the portal. Time spent by 25 randomly chosen visitors was measured and the results of these measurements are presented below:

31.90	34.47	23.06	33.95	43.38
30.53	41.20	29.98	29.32	17.40
44.82	43.21	19.46	39.32	24.72
42.86	29.58	36.46	20.70	28.19
27.92	40.15	30.59	19.15	25.13

- a) Give, based on the relative frequency, an estimate for the probability that a randomly selected visitor will spend more than 30 seconds on the main page of the portal.

The executives would like the population distribution of the amount of time spent on the main page to be normally distributed with mean $\mu = 30.00$ seconds and standard deviation $\sigma = 10.00$ seconds.

- b) Give, assuming that the goal of the executives is achieved, the probability that a randomly selected visitor will spend less than 28 seconds on the main page of the portal.
- c) Give, assuming that the goal of the executives is achieved, the probability that the mean time spent on the main page of the portal by 36 randomly selected visitors is less than 28 seconds.
3. Every employee of a big company received recently a new laptop. All laptops come with a pre-installed operating system, including, among others, a highly unstable web browser Verge. It turns out that the probability of Verge crashing at least once while surfing the Internet equals 0.80. Since Verge has such bad reputation, many users choose to replace it with a more stable web browser called Marble. Some time after the introduction of the new laptops the IT department of the company surveyed its employees about the user experience. It turns out that 92% of employees who experienced at least one crash of Verge installed Marble, and 46% of employees who did not have any problems with Verge decided to switch to Marble anyway.

- a) Compute the probability that a randomly chosen employee has Marble installed on their laptop.
- b) A randomly chosen employee has Marble installed on their laptop. Compute the probability that Verge, installed previously on their laptop, crashed at least once. Round the result to three decimal places. (To one decimal place, if you express probabilities in percentages.)
- c) Do crashes of Verge increase the probability of installing Marbles? Does it imply that the events $A = \{\text{Verge crashed at least once}\}$ and $B = \{\text{Marble is installed}\}$ are dependent? Why (not)?

4. Andy first rolls a fair six-sided die, and then tosses a fair coin.
- Give the sample space Ω and probability measure P for this experiment.
 - How would the probability measure P change if Andy used a biased coin with the probability of *Tails* equal to 0.4?

Andy proposes the following game (with a fair six-sided die and a fair coin) to Bridget: if *Heads* comes up, Bridget receives 3 euros from Andy, irrespective of the result on the die. If *Tails* comes up, Bridget has to pay Andy as many euros as the die shows.

- Let X be the random variable which denotes the amount Bridget earns in one turn of the game. Determine all possible values x of X and construct the probability function $p(x) = P(X = x)$ of X based on the formal definition. (You may present the results in a table).
 - Compute, using part c), the expectation $E(X)$ of X .
5. Figure 1 shows histograms of Sample A and Sample B , a boxplot of Sample B , and three normal QQ-plots. It is known that Samples A and B originate from two different population distributions.
- By looking at both histograms, what can be said about the tails of the population distribution of Sample A with respect to the tails of the population distribution of Sample B ?
 - What does the box (i.e., the three lines: top and bottom sides of the box and the thick line in the middle) in a boxplot represent?
 - Two out of the three normal QQ-plots correspond to Sample A and Sample B . Indicate which QQ-plot depicts Sample A and which one depicts Sample B , and motivate your answer.

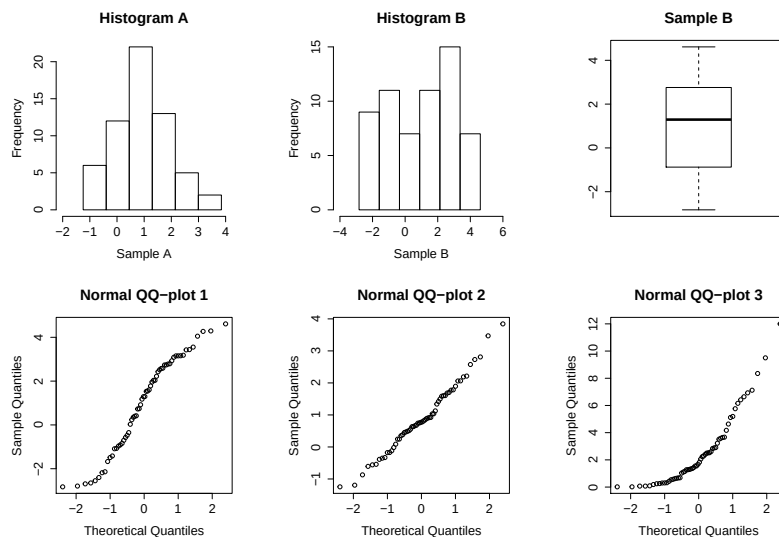


Figure 1: Histograms of Sample A and Sample B , boxplot of Sample B , three QQ-plots.

Formulas and Tables for Exam Statistical Methods

Probability

We use the following notation:

Ω sample space, P probability measure.

$B, A_1, A_2, \dots, A_m$ events,

$A_1, A_2, \dots, A_m$ a partition of Ω with $P(A_i) > 0$ for all $i \in \{1, 2, \dots, m\}$.

Law of Total Probability:

$$P(B) = \sum_{i=1}^m P(B \cap A_i) = \sum_{i=1}^m P(B|A_i)P(A_i).$$

Bayes' Theorem:

$$P(A_r|B) = \frac{P(A_r \cap B)}{\sum_{i=1}^m P(B|A_i)P(A_i)} = \frac{P(B|A_r)P(A_r)}{\sum_{i=1}^m P(B|A_i)P(A_i)}.$$

NEGATIVE z Scores

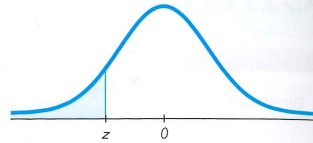


Table 2 Standard Normal (z) Distribution: Cumulative Area from the LEFT

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.50 and lower	.0001									
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

NOTE: For values of z below 3.49, use 0.0001 for the area.

*Use these common values that result from interpolation:

z Score	Area
-1.645	0.0500
-2.575	0.0050

(continued)

POSITIVE z Scores

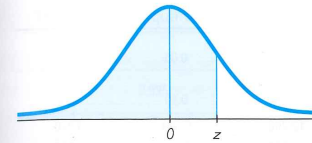


Table 2 (continued) Cumulative Area from the LEFT

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998
3.50 and up	.9999									

NOTE: For values of z above 3.49, use 0.9999 for the area.

*Use these common values that result from interpolation:

z Score	Area
1.645	0.9500
2.575	0.9950

Common Critical Values

Confidence Level	Critical Value
0.90	1.645
0.95	1.96
0.99	2.575